Canvas3D: Object-Level 3D Lifting and Line-Art Composition for Painting Animation

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Abstract

Animating static artworks—oil paintings, ink-wash scrolls, and frescoes—into explorable, narratively coherent videos is a long-standing goal at the intersection of computer vision, generative modeling, and digital cultural heritage. Existing approaches remain bifurcated: image-to-video diffusion models produce visually fluent motion but collapse three-dimensional structure under viewpoint change, while feed-forward 3D reconstruction pipelines capture geometry at the cost of obliterating painterly texture, impasto, and brushstroke identity. The root cause is a fundamental *objective tension*: artistic style fidelity and 3D scene structure are governed by incompatible optimization targets, and no single end-to-end model satisfies both, especially when the task demands object-level manipulation. We propose **Canvas3D**, a framework that resolves this tension through a *modular decomposition of concerns*. Rather than training a monolithic network, Canvas3D factors the painting-to-video problem into four independently optimized, mutually reinforcing sub-problems: *(i) perceptual grouping* of a painting into semantically meaningful foreground objects and a preserved background, jointly with monocular depth-derived layer ordering; *(ii) per-object geometric lifting* that converts each 2D region into a metric-scaled 3D representation and assembles a reusable, retrievable object-level model library; *(iii) line-art scene composition* in which hand-drawn sketches serve as *spatial constraints* rather than stylistic hints, solved by a contour-based rigid registration that places each 3D object at its sketched location and a depth-conditioned inpainting module that restores disoccluded background texture in a geometry-aware manner; and *(iv) geometry-grounded video synthesis* that renders the composited 3D scene into multi-view keyframes and injects depth and edge cues as dual control signals into a video diffusion backbone, anchoring temporal motion to explicit 3D structure rather than learned heuristics. The key insight is that *separating perceptual, geometric, compositional, and temporal concerns* turns an ill-posed, conflicting joint problem into four well-posed sub-problems, each addressable by an off-the-shelf specialist, while their interaction yields capabilities—arbitrary object repositioning, multi-view explorability, and style-anchored animation—that no single-modality pipeline offers. On a curated benchmark of classical paintings, Canvas3D achieves state-of-the-art style fidelity, perceptual quality, and temporal coherence, with user studies confirming strong preference for artistic authenticity, motion naturalness, and compositional-intent alignment.

1 Introduction

Classical paintings—whether oil on canvas, ink on silk, or pigment on fresco—are cultural artifacts that encode centuries of human visual imagination Lowenthal [2015]. From van Gogh's swirling *Starry Night* to Zhang Zeduan's panoramic *Along the River During the Qingming Festival*, these works imply motion, narrative, and spatial depth that their static medium can only suggest. Animating such paintings as *explorable* videos—sequences in which the viewer's viewpoint can move, and in which painted subjects can be repositioned according to the animator's compositional intent—has therefore emerged as a compelling research goal at the intersection of computer vision, generative modeling, and digital cultural heritage Zhang *et al.* [2024]; Dong [2024].

Recent years have witnessed remarkable progress on two fronts that, however, fail to meet on common ground. On one hand, image-to-video (I2V) diffusion models Blattmann *et al.* [2023]; Wan Team [2025]; Lightricks Team [2025]; Yang *et al.* [2025], combined with controllable conditioning mechanisms Zhang *et al.* [2023]; Wang *et al.* [2024c]; Peng *et al.* [2024], can animate a single photograph into a short clip with plausible per-frame dynamics. Yet when applied directly to oil paintings or ink-wash artworks, they exhibit two fun-

damental shortcomings. First, they operate entirely in 2D pixel space: the model has no geometric understanding of object depth ordering, self-occlusion, or relative spatial layout, and consequently can neither synthesize novel viewpoints nor maintain consistent 3D structure across frames. Second, long-range temporal coherence degrades rapidly beyond 2–4 seconds, as latent-space interpolation lacks an explicit geometric constraint to anchor the motion trajectory.

On the other hand, 3D reconstruction and rendering methods Mildenhall *et al.* [2020]; Kerbl *et al.* [2023]; Huang *et al.* [2024] can reconstruct explorable 3D scenes from sparse views, and feed-forward monocular variants Wang *et al.* [2024e]; Tang *et al.* [2024]; Charatan *et al.* [2024] can even lift a single image to a 3D representation. However, these reconstruction pipelines are optimized for photorealistic fidelity; when applied to paintings, the rendering process replaces expressive brushstrokes with geometric artifacts, flattens the impasto texture, and bleeds semantic color boundaries under novel-view projection.

The core tension is this: *3D scene structure and artistic style fidelity* are governed by incompatible optimization objectives, and no single model family—neither pure diffusion nor pure 3D reconstruction—satisfactorily achieves both. The tension becomes especially acute when the animation task demands **object-level manipulation**: moving a painted figure from one compositional position to another while preserving its painterly identity and filling the resulting disocclusions with plausible background content.

Our approach: modular separation of concerns. Rather than training a single monolithic network to jointly optimize style and structure—an ill-posed objective whose gradients pull in opposite directions—we propose to *factor* the painting-to-video problem into four interacting yet independently optimized sub-problems, each addressed by a specialist and communicating with the next through a small, interpretable handshake representation (Figure 1):

- **PaintParse** handles *perceptual grouping*: decomposing the input painting into foreground objects and a background layer using an art-domain-adapted segmentation backbone, augmented with monocular depth estimation for metric-scale layer ordering.
- **MonoSplat** handles *geometric structure*: lifting each segmented foreground region into a high-fidelity 3D representation via feed-forward single-image 3D prediction, assembling an object-level model library with depth-derived metric rescaling.
- **SketchComp** handles *compositional intent*: accepting hand-drawn line-art sketches specifying target object positions, solving a contour-based rigid registration to place each 3D object onto its sketch, and filling disocclusion gaps through a depth-conditioned inpainting module that preserves background artistic texture.
- **ArtiFlow** handles *temporal rendering*: rendering the composited 3D scene into multi-view keyframes and feeding them into a video diffusion backbone with dual control branches (depth + edge), producing temporally smooth animations with pixel-precise object motion governed by explicit 3D geometry rather than learned heuristics.

This decomposition enables a clean separation of concerns: each handshake carries exactly the information the downstream stage requires and nothing more, which localizes errors instead of compounding them. The result is a system that supports *arbitrary object repositioning*, *multi-view explorability*, and *style-anchored animation*—capabilities that no prior single-modality pipeline offers.

Contributions. Our contributions are:

1. **A modular factorization** of the painting-to-video problem into four independently optimized specialists—PaintParse (perceptual grouping), MonoSplat (per-object 3D lifting), SketchComp (line-art composition with depth-aware inpainting), and ArtiFlow (dual-conditioned video synthesis)—that communicate through small, interpretable handshake representations. To our knowledge, this is the first end-to-end pipeline for 3D-aware, artistically controllable painting animation.
2. **An object-level 3D lifting pipeline** (MonoSplat) that couples art-domain-adapted instance segmentation with feed-forward monocular 3D prediction and depth-derived metric rescaling (Eq. 2), yielding a reusable, retrievable library of metric-consistent 3D foreground objects from a single painting.
3. **A sketch-conditioned 3D scene composition algorithm** (SketchComp) whose contour-based rigid registration jointly optimizes region IoU and boundary Chamfer distance (Eq. 3), complemented by a depth-conditioned inpainting module that fills disocclusion gaps (Eq. 4) while preserving the background's artistic texture.
4. **A geometry-grounded video synthesis workflow** (ArtiFlow) whose dual control branches (depth + edge, Eq. 5) and first-frame latent anchoring anchor temporal motion to explicit 3D structure with per-object control.
5. **Comprehensive evaluation** on a curated benchmark of 50 classical oil paintings and ink-wash works, achieving state-of-the-art style fidelity, perceptual quality, and temporal coherence, with user studies confirming strong preference for artistic authenticity, motion naturalness, and compositional-intent alignment.

2 Related Work

We organize prior work along the four axes that motivate our modular decomposition, tracing the developmental lineage of each thread and positioning Canvas3D at their intersection.

2.1 From Pixel-Level Stylization to Painting Animation

The earliest approaches to painting animation treat the input as a *pixel grid to be stylized over time*. Static image style transfer, pioneered by feature-correlation methods **?** and extended by fast feed-forward networks **??** and transformer-based stylizers **?**, established that artistic appearance can be re-injected into arbitrary content. Video style transfer then addressed temporal coherence through optical-flow regularization **???**, multi-channel correlation **?**, and content deformation fields **?**. A second lineage pursues *stroke-level synthesis*—reproducing the physical painting process as a time-lapse **?**Zou *et al.* [2021]; Liu *et al.* [2021]—but such methods reconstruct *how a painting was made* rather than *how its subjects could move*. A third, narrower thread drives *portrait motion* on painted faces Guo *et al.* [2024]; Tan *et al.* [2024] but cannot animate full-body figures or background elements. The diffusion era introduced inversion-based stylization **?** and video-consistent editing **????**, which preserve appearance under text-driven edits but provide no geometric handle on object pose. The recent InkSpirit **?** and StyleMaster **?** push appearance fidelity further yet remain confined to 2D image translation. Across this lineage, the persistent gap is the absence of *object-level, 3D-grounded motion control*: appearance is preserved, but spatial structure is not.

2.2 3D Scene Representations and Single-Image Lifting

Three-dimensional representations evolved from implicit radiance fields Mildenhall *et al.* [2020] to explicit primitives. 3D Gaussian Splatting Kerbl *et al.* [2023] represents a scene as a set of anisotropic Gaussians with differentiable rasterization, enabling real-time photorealistic rendering; 2D Gaussian Splatting Huang *et al.* [2024] collapses the third scale dimension into a surface normal, improving geometric fidelity for surface-like content. Crucially, the field has recently shifted from *per-scene optimization* to *feed-forward single-image lifting*: generative-prior methods Tang *et al.* [2024] predict 3D assets from one image, while state-of-the-art feed-forward predictors Wang *et al.* [2024e]; Charatan *et al.* [2024]; Jiang *et al.* [2025] regress Gaussian parameters directly, with two-view and unconstrained-view variants broadening the input regime. These advances make it feasible to lift individual painting regions into 3D, but none addresses the downstream challenge of *composing multiple such objects into a coherent, artistically controllable scene*—exactly the gap our framework targets.

2.3 Sketch and Line-Art as a Composition Interface

Hand-drawn sketches have a long history as an intuitive interface for shape prototyping Olsen *et al.* [2009]. Recent work demonstrates that contour-level spatial reasoning is tractable with modern vision models: ShadowDraw Luo *et al.* [2026] generates compositional shadow art from object inputs, LineArt Wang *et al.* [2024d] performs training-free appearance transfer onto design drawings via diffusion, and region-aware simplification Nguyen *et al.* [2024] and volumetric animation toolkits Fox-Gieg [2024] illustrate the expressive potential of hand-drawn control. In diffusion-based generation, sketches typically serve as a *stylistic* or *boundary* condition (e.g., ControlNet-style edge injection Zhang *et al.* [2023]). Canvas3D departs from this convention by treating line-art as a *spatial constraint* that resolves the which-object-where problem in 3D scene assembly: a sketch is not merely an appearance hint but a target silhouette that a 3D object is geometrically registered to.

2.4 Controllable Image-to-Video Generation

The leap from text-to-image to text-to-video was powered by diffusion transformers Peebles and Xie [2023] and video diffusion models Ho *et al.* [2022]; **?**. A family of scalable backbones Yang *et al.* [2025]; Wan Team [2025]; Lightricks Team [2025]; Blattmann *et al.* [2023] now supports high-fidelity image-to-video synthesis. Controllability has been pursued through auxiliary conditioning: ControlNet Zhang *et al.* [2023] injects edge, depth, and pose maps into the diffusion U-Net, with video extensions Wang *et al.* [2024c] and efficiency-oriented variants Peng *et al.* [2024] broadening its applicability. Cinematic shot design Xing *et al.* [2025] and clipart animation Wu *et al.* [2025] demonstrate compelling 3D-aware control without explicit 3D supervision. Despite these advances, existing I2V control methods are confined to *per-frame conditioning signals*: they cannot reason about 3D object identity, depth ordering, or occlusion across frames, and consequently drift under viewpoint change. Canvas3D addresses this by grounding all control signals in an explicit, object-level 3D scene that is composited, rendered, and then fed into the video diffusion backbone.

2.5 Painting-Specific Animation Pipelines

A small but growing body of work targets paintings directly. Domain-adapted detectors Chen *et al.* [2025] and art-segmentation datasets Saleh and Elgammal [2016] establish that natural-image segmentation backbones Jocher *et al.* [2023]; Wang *et al.* [2024b,a]; Tian *et al.* [2025] and foundation segmentation models Kirillov *et al.* [2023]; Ravi *et al.* [2024] require art-specific fine-tuning due to brushstroke texture and intentional boundary blending. Chinese ink-wash style transfer has been explored via instance-aware coherence **?** and staged diffusion guidance **?**. Most relevant to our setting, "Printing the Underdetermined" Ming *et al.* [2026] translates paintings to 3D representations via video generative models but operates on the entire image jointly rather than at the object level, while OilPaintingRevive Anonymous [2026] proposes a multi-stage pipeline combining 2DGS with video refinement, but its skeleton-binding network requires per-scene optimization that limits compositional flexibility. Canvas3D advances beyond these

works along four axes—*object-level* rather than image-level manipulation, *line-art* as a compositional control interface, *depth-aware inpainting* for disocclusion, and *multi-modal geometric conditioning* for video synthesis—yielding stronger 3D consistency and explicit compositional control.

3 Method

Design philosophy. A single painting-animation system must simultaneously satisfy four *conflicting* objectives: perceptual grouping of artistically rendered content, geometric reconstruction of objects that have no real 3D extent, user-specifiable composition, and temporally coherent stylized rendering. Jointly optimizing a single network for all four is ill-posed because the gradients of style fidelity and 3D structure pull in opposite directions. Our central design principle is therefore *modular separation of concerns with explicit handshakes*: each objective is addressed by an independently optimized specialist, and the four specialists communicate only through well-defined, interpretable intermediate representations—masks with depth-ordered layers, a metric-scaled object-level 3D library, a composited 3D scene with disocclusion-filled background, and finally a multi-view keyframe stream with paired geometry and edge cues. This decomposition converts a conflicting joint problem into four well-posed sub-problems, while the handshakes carry exactly the information each downstream stage needs and no more, preventing error compounding.

Problem statement. Given a painting scan $P \in \mathbb{R}^{H\times W\times 3}$ and a set of hand-drawn line-art sketches $L = \{\ell_1, \ldots, \ell_K\}$ specifying target object positions, Canvas3D produces an animated video V that (i) preserves the artistic style of P, (ii) supports viewpoint exploration of a 3D-composited scene, and (iii) places each foreground object at its sketched location with geometrically grounded, temporally coherent motion. The framework comprises four modules (Figure 1): **PaintParse** (perceptual grouping and layer decomposition), **MonoSplat** (per-object 3D lifting and library assembly), **SketchComp** (line-art-guided scene composition with depth-aware inpainting), and **ArtiFlow** (geometry-grounded video synthesis).

Notation. Let $\Omega = \{1, \ldots, H\} \times \{1, \ldots, W\}$ denote the pixel grid of the painting. For the i-th foreground object ($i \in \{1, \ldots, K\}$), $M_i \in \{0,1\}^{H\times W}$ is its binary mask and $\Omega_i = \{p \in \Omega \mid M_i(p) = 1\}$ the associated pixel set; the residual background mask is $M_{\text{bg}} \in \{0,1\}^{H\times W}$. The (metric-aligned) monocular depth map is $D_{\text{mono}} : \Omega \to \mathbb{R}_+$; each object is summarized by its *median* depth $d_i = \text{median}\{D_{\text{mono}}(p) \mid p \in \Omega_i\}$, used for metric recovery, together with a robust depth range $[d_i^{\text{near}}, d_i^{\text{far}}]$ (the 10th and 90th percentiles over Ω_i), used only for layer ordering. The 3D representation of object i is $\mathcal{G}_i$ and the assembled library is $\{\mathcal{G}_1, \ldots, \mathcal{G}_K\}$; the composited scene and its depth map are $\mathcal{S}^*$ and $\mathcal{D}^*$; Ω_{gap} is the disocclusion region; and $\{I_t, D_t, E_t\}_{t=1}^T$ are the multi-view RGB, depth, and edge keyframes rendered along camera poses $\{\mathbf{T}_t\}_{t=1}^T$. The focal length in pixels is f. Throughout, $\pi(\cdot)$ denotes the perspective projection onto the image plane and $\text{IoU}(\cdot,\cdot)$ the intersection-over-union between two binary regions.

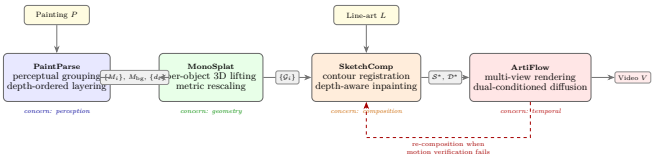


Figure 1: **Canvas3D overview.** The pipeline factors painting animation into four independently optimized specialists—*PaintParse* (perceptual grouping and depth-ordered layering), *MonoSplat* (per-object 3D lifting into a reusable, metric-scaled library), *SketchComp* (line-art-guided scene composition with depth-aware inpainting), and *ArtiFlow* (geometry-grounded video synthesis). Adjacent specialists communicate exclusively through small, interpretable *handshake representations* (gray tokens): depth-ordered masks, a metric 3D object library, and the composited scene with its depth map. The dashed loop re-enters SketchComp when ArtiFlow's per-object motion verification fails.

3.1 PaintParse: Art-Domain Perceptual Grouping and Layer Ordering

Design rationale. The downstream 3D and composition stages require two distinct outputs that no single head of an off-the-shelf detector provides: *(i)* foreground masks whose boundaries respect artistic intent rather than photographic edges, and *(ii)* a depth-consistent layer ordering that disambiguates which object lies in front of which. PaintParse therefore couples a domain-adapted instance segmentation backbone with a parallel monocular depth estimator and fuses their outputs through a percentile-based layer assignment. The key design choice is *decoupling semantic grouping from depth ordering*: the segmentation network is responsible only for "what is an object," while the depth estimator is responsible only for "how far away it is," and the layer-assignment logic simply reads both off. This avoids asking either network to solve a problem it was not trained for.

Instance segmentation with art-domain adaptation. We adopt a one-stage instance segmentation backbone pre-trained on natural-image data. Direct transfer to paintings yields poor mask quality because brushstroke textures produce spurious edge responses and depth-ambiguous compositions confound object–background separation. We mitigate this through a two-stage adaptation protocol: a *low-rank adaptation* of the detection backbone, which injects art-domain statistics while constraining the parameter drift that would cause catastrophic forgetting, followed by *full fine-tuning* of the mask-prediction head to align boundary prediction with painterly edges. Given input P, the adapted model outputs K binary masks $\{M_i\}_{i=1}^K$, ordered such that $\Omega_i \cap \Omega_j = \varnothing$ for $i \neq j$ (overlapping detections are sup-

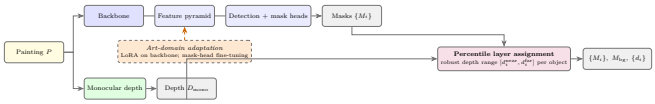


Figure 2: **PaintParse.** A segmentation branch (top) and a monocular depth branch (bottom) run in parallel on the input painting. Art-domain adaptation (dashed) injects painting-specific statistics into the backbone and sharpens the mask head. A percentile-based layer assignment fuses both outputs into depth-ordered layers, treating each object as an atomic unit.

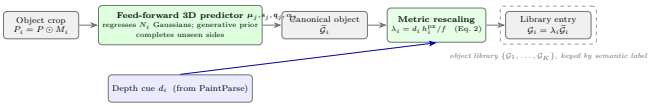


Figure 3: **MonoSplat.** Each foreground crop is lifted in a single feed-forward pass into a canonical-space Gaussian representation, whose generative prior completes the object sides absent from the painting. The PaintParse depth cue d_i restores metric scale (Eq. 2), placing every object in a common metric frame so that the resulting library entries can be composed across paintings.

pressed by non-maximum suppression and mask deduplication). The residual background mask is the complement of the foreground union:

$$M_{\text{bg}}(p) \;=\; 1 \;-\; \max_{i\in\{1,\ldots,K\}} M_i(p), \qquad \forall p \in \Omega. \quad (1)$$

Monocular depth and layer assignment. In parallel, a monocular depth estimator produces a relative depth map $D_{\text{mono}} : \Omega \to \mathbb{R}_+$ (nearness-coded: smaller values indicate closer content). For each detected object i we summarize its depth by the 10th and 90th percentiles of $\{D_{\text{mono}}(p) \mid p \in \Omega_i\}$, yielding the robust range $[d_i^{\text{near}}, d_i^{\text{far}}]$ that is insensitive to outlier pixels at mask boundaries, and by the median $d_i = \text{median}\{D_{\text{mono}}(p) \mid p \in \Omega_i\}$, which the MonoSplat stage consumes for metric recovery. The layer index of each object is determined by the depth percentile of its mask centroid, i.e., object i is assigned a *lower* (front) layer than object j whenever the centroid depth of i is smaller. This percentile-based assignment is preferable to a per-pixel depth sort because it treats each object as an atomic unit, preserving its identity under the rigid transformations that subsequent composition applies.

3.2 MonoSplat: Per-Object 3D Lifting and Library Assembly

Design rationale. A whole-scene 3D reconstruction would entangle object geometry with background texture, making independent repositioning impossible. We therefore lift each foreground object *separately* into a 3D representation and assemble a *reusable library*, indexed by semantic label and original painting coordinates. Two design choices follow from this principle. First, we use a *feed-forward single-image 3D predictor* rather than per-scene optimization, because per-object optimization would be prohibitively expensive and would overfit to a single viewpoint. Second, because feed-forward predictors operate in a normalized canonical space without metric scale, we *recover metric dimensions from the PaintParse depth cue*, so that objects from different paintings can later be composed in a common metric frame.

Feed-forward lifting. For each foreground object i, we crop the painting to its mask, $P_i = P \odot M_i$, and feed the crop into a feed-forward single-image 3D predictor that regresses a set of N_i anisotropic 3D Gaussians directly from one image. Each Gaussian j is parameterized by a center $\boldsymbol{\mu}_j \in \mathbb{R}^3$, scales $\mathbf{s}_j \in \mathbb{R}^3_+$, rotation unit quaternion $\mathbf{q}_j \in \mathbb{S}^3 \subset \mathbb{R}^4$, opacity $\alpha_j \in [0,1]$, and view-dependent color coefficients, and the object is rendered by α-composited Gaussian splatting. The predictor's generative prior, trained on diverse 3D assets, generalizes to the missing side and back of objects that are not visible in the painting—a critical capability for the rotations that downstream composition imposes.

Metric rescaling. The predictor outputs geometry in a normalized canonical space, i.e., it estimates shape $\widehat{\mathcal{G}}_i$ up to an unknown similarity transform. To place objects from different paintings (and different object classes) in a single metric frame, we recover the scale factor from the pinhole imaging relation. Under the standard thin-lens approximation, the physical height h_i of an object, its image-plane extent h_i^{px} in pixels, its depth d_i, and the focal length f (in pixels) satisfy $h_i = d_i\, h_i^{\text{px}}/f$. We therefore rescale the canonical geometry by

$$\lambda_i \;=\; \frac{d_i \cdot h_i^{\text{px}}}{f \cdot h_i^{\text{can}}}, \quad (2)$$

where d_i is the median depth of object i (from PaintParse), h_i^{px} is its mask height in the painting, and h_i^{can} is the height of its bounding volume in the predictor's canonical output space; the ratio $h_i^{\text{px}}/h_i^{\text{can}}$ converts the canonical unit to pixels before the depth relation maps it to metric units. Each Gaussian center and scale is rescaled as $\boldsymbol{\mu}_j \mapsto \lambda_i \boldsymbol{\mu}_j$, $\mathbf{s}_j \mapsto \lambda_i \mathbf{s}_j$, producing a metric-consistent 3D object $\mathcal{G}_i = \lambda_i \widehat{\mathcal{G}}_i$. The resulting library $\{\mathcal{G}_1, \ldots, \mathcal{G}_K\}$ is keyed by semantic label and original painting coordinates, enabling selective retrieval and cross-painting composition in future work.

3.3 SketchComp: Line-Art-Guided 3D Scene Composition

Design rationale. The user's compositional intent—which object goes where—is most naturally expressed as a sketch, but existing diffusion-based sketch control treats sketches as appearance or boundary cues rather than spatial constraints. SketchComp inverts this convention: a sketch *is* the target, and a 3D object is *geometrically registered* to it. Two sub-problems must then be solved: *(i)* finding the rigid transformation that aligns each 3D object's rendered silhouette with its sketched

contour, and *(ii)* filling the disocclusion gaps that open up when an object moves away from its original position, while preserving the background's artistic texture. We solve the first with a silhouette-overlap objective that combines region IoU with boundary Chamfer distance, and the second with a depth-conditioned inpainting module whose geometric conditioning prevents the texture-smear artifact that naive inpainting produces on paintings.

Contour-based rigid registration. Each sketch ℓ_i is converted to a binary edge map via Canny detection, and its external contour $\Gamma_i \subset \Omega$ is extracted. For each 3D object $\mathcal{G}_i$, we search for the rigid transformation—in-plane rotation $\theta \in [0, 2\pi)$, 2D translation $\mathbf{t} \in \mathbb{R}^2$, and uniform scale $s \in \mathbb{R}_+$—that best aligns the rendered silhouette with Γ_i:

$$(\theta_i^*, \mathbf{t}_i^*, s_i^*) = \arg\min_{\theta,\,\mathbf{t},\,s} \mathcal{L}_{\text{cont}}(\theta, \mathbf{t}, s), \tag{3}$$

$$\mathcal{L}_{\text{cont}}(\theta, \mathbf{t}, s) = \underbrace{1 - \text{IoU}\big(R_i(\theta, \mathbf{t}, s),\, R(\Gamma_i)\big)}_{\text{region}} + \lambda_c \underbrace{\text{CD}\big(\partial R_i(\theta, \mathbf{t}, s),\, \Gamma_i\big)}_{\text{boundary}},$$

where $R_i(\theta, \mathbf{t}, s)$ is the binary silhouette obtained by rendering $\mathcal{G}_i$ from the canonical camera under the candidate transform, $R(\Gamma_i)$ is the filled region enclosed by the sketched contour, ∂R_i denotes its boundary curve, $\text{CD}(A, B) = \frac{1}{|A|}\sum_{a \in A} \min_{b \in B} \|a - b\|_2$ is the directed Chamfer distance (in pixels), and $\lambda_c > 0$ balances the two terms. The IoU term rewards global region agreement, while the Chamfer term penalizes boundary misalignment; their combination avoids the degenerate solutions that either term alone admits (a small silhouette placed inside the contour attains zero Chamfer error at the wrong scale, while region-only matching tolerates topologically incorrect regions whose interiors overlap). Optimization proceeds by gradient descent with iterative re-rendering, so that each step's gradient is computed against the current geometry rather than a stale approximation; the resulting transform is applied to $\mathcal{G}_i$ to obtain the placed object $\mathcal{G}_i'$.

Scene assembly and disocclusion inpainting. Each registered object is placed at its PaintParse-derived depth d_i, and objects are rendered in depth-sorted (painter's) order against a background layer reconstructed from the RGB-D map. Object displacement creates *disocclusion gaps*: background pixels that become newly visible because the surface content previously occluded by the object in its original position is not recorded in the single input image. We detect the gap as the set of pixels inside the original foreground support whose projected scene content is missing:

$$\Omega_{\text{gap}} \;=\; \Big\{\, p \in \Omega \;\Big|\; \mathcal{V}_{\mathcal{S}}(p) = 0 \Big\}, \tag{4}$$

where $\mathcal{V}_{\mathcal{S}}(p) = 1$ if and only if at least one 3D surface in the assembled scene $\mathcal{S}$ projects onto pixel p with valid

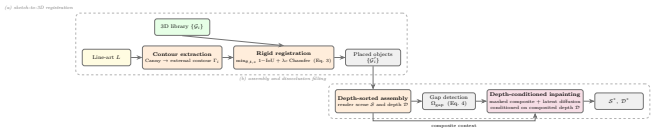


Figure 4: **SketchComp.** (a) Hand-drawn contours are matched to rendered 3D silhouettes by a rigid registration that jointly optimizes region IoU and boundary Chamfer distance (Eq. 3). (b) Registered objects are assembled in depth-sorted order; the disocclusion gaps opened by object displacement (Eq. 4) are filled by a depth-conditioned inpainting diffusion module that respects the background's geometric structure, yielding the composited scene $\mathcal{S}^*$ and its depth map $\mathcal{D}^*$.

appearance (in practice, pixels covered by the original M_{bg} support or by any placed object). Filling Ω_{gap} while preserving artistic texture coherence is critical. We employ a depth-conditioned inpainting diffusion pipeline: a masked composite (with Ω_{gap} zeroed and dilated to expose surrounding context as texture reference) is fed into a latent inpainting diffusion model, with a depth-conditioned control branch conditioned on the composited depth map $\mathcal{D}$. The depth conditioning is the key design choice: it constrains the inpainted content to follow the background's geometric structure, preventing the texture-smear artifact that arises when an unconstrained inpainter fills a painting gap with semantically plausible but geometrically inconsistent content. The resulting scene $\mathcal{S}^*$ and its depth map $\mathcal{D}^*$ are forwarded to ArtiFlow.

3.4 ArtiFlow: Geometry-Grounded Video Synthesis

Design rationale. Pure image-to-video diffusion applied to a painting produces plausible per-frame motion but suffers two failure modes: geometric drift (objects wander off their intended trajectories) and boundary melting (brushstroke edges blur across frames). ArtiFlow addresses both by *grounding the diffusion in explicit 3D geometry*: the composited scene is rendered into a multi-view keyframe stream that provides three control signals—RGB appearance, depth geometry, and edge boundary—which are injected into a video diffusion backbone as dual control branches. The depth branch anchors each frame to the 3D structure of $\mathcal{S}^*$, suppressing geometric drift; the edge branch preserves brushstroke-level boundary detail, suppressing boundary melting. The combined conditioning converts the diffusion backbone from a free-form motion generator into a *geometry-anchored refiner* whose temporal freedom is restricted to the residual motion that the 3D rendering does not already specify.

Multi-view keyframe rendering. Given $\mathcal{S}^*$, we define a smooth camera trajectory through T poses $\{\mathbf{T}_t\}_{t=1}^{T} \subset SE(3)$, generated by cubic B-spline interpolation between a canonical front-facing start pose and a user-specified or auto-generated end pose. At each step t, differentiable rasterization produces an RGB frame

$I_t = \mathcal{R}_{\text{rgb}}(\mathcal{S}^*, \mathbf{T}_t)$, a depth map $D_t = \mathcal{R}_{\text{depth}}(\mathcal{S}^*, \mathbf{T}_t)$, and an edge map $E_t = \text{Edge}(I_t)$. The keyframe count is set to $T = 8N + 1$ for some $N \in \mathbb{N}$, aligning the keyframe stream with the temporal chunking of modern video diffusion backbones.

Dual-conditioned video diffusion. Two parallel control branches inject geometric and boundary cues into the video diffusion backbone. The depth branch, conditioned on $\{D_t\}$ at strength $s_d \in [0, 1]$, anchors each frame to the 3D structure of $\mathcal{S}^*$, preventing the geometric drift that plagues pure I2V models. The edge branch, conditioned on $\{E_t\}$ at strength $s_e \in [0, 1]$, preserves brushstroke-level boundary detail, suppressing the boundary-melting artifact common when diffusion is applied to paintings. Writing $\mathbf{c}_t^{\text{depth}}$ and $\mathbf{c}_t^{\text{edge}}$ for the control features produced by the two branches at frame t, the combined conditioning signal is

$$\mathbf{c}_t \;=\; s_d\, \mathbf{c}_t^{\text{depth}} \;+\; s_e\, \mathbf{c}_t^{\text{edge}}, \qquad t = 1, \ldots, T, \qquad (5)$$

injected at the corresponding residual connections of the backbone. The two feature streams share a common embedding space (both control encoders project into the backbone's residual channels), so the weighted sum in Eq. 5 is well defined. Temporal coherence is further reinforced by *first-frame latent anchoring*: the initial latent of frame $t > 1$ is set to the encoded first frame plus a position-encoded Gaussian perturbation $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \sigma_t^2 \mathbf{I})$ with $\sigma_t = \sigma_{\max}\,(t-1)/(T-1)$ growing linearly in the temporal distance $|t-1|$, so that motion freedom increases smoothly away from the anchored first frame while the drift budget remains bounded by $\sigma_{\max}$.

Per-object motion verification. After generation, we verify motion fidelity by computing the IoU between rendered object masks and segmented masks in each generated frame. If the average IoU falls below a threshold τ_{mo}, the pipeline triggers a *re-composition* step (dashed arrow in Figure 1) that adjusts control strengths or regenerates the affected segment with stronger depth conditioning, closing the loop between composition and synthesis.

3.5 Pipeline Algorithm and Implementation

Algorithm 1 summarizes the full Canvas3D pipeline. The modular design enables *fault-tolerant resumption*: each stage saves its outputs as checkpoint files, so any specialist can be upgraded or replaced without re-running downstream stages—for example, swapping in a stronger 3D predictor or a different video backbone. Peak memory consumption across all four stages is kept within a single 24 GB budget through serial model loading with explicit memory release between stages, which is feasible precisely because the handshake representations between stages are small, serializable tensors rather than shared activations.

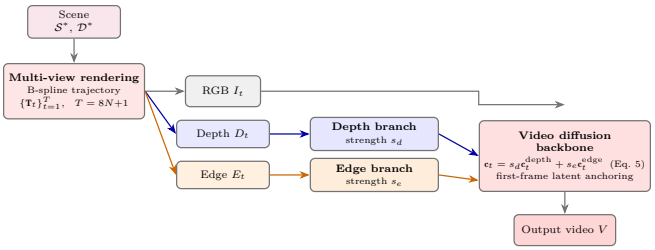


Figure 5: **ArtiFlow.** The composited 3D scene is rendered along a smooth B-spline camera trajectory into paired RGB, depth, and edge keyframes. Depth and edge signals enter the video diffusion backbone through two control branches whose combined conditioning (Eq. 5) anchors every frame to the scene's 3D geometry, while first-frame latent anchoring restricts temporal freedom to the residual motion not already specified by the rendering.

4 Experiments

We structure the experimental evaluation around four research questions: **Q1** Does Canvas3D outperform state-of-the-art image-to-video, 3D reconstruction, and painting-animation methods on a curated painting benchmark? **Q2** How much does each of the four modules contribute to the final quality, and which design choices within each module are load-bearing? **Q3** Do human viewers prefer Canvas3D outputs on artistic authenticity, motion naturalness, and compositional-intent alignment? **Q4** How does the runtime and memory budget decompose across modules, and where does the computational cost concentrate?

4.1 Experimental Setup

Dataset. We curate the **PaintingAnim-50** benchmark comprising 50 high-resolution digital scans of classical paintings (30 oil, 15 ink-wash, 5 fresco) drawn from both Western and East Asian traditions. The collection spans diverse compositional types: single-figure portraits (12), multi-figure narrative scenes (18), landscapes with architectural elements (12), and animal or subject-centered works (8). Each painting is annotated with instance-level segmentation masks (5–15 objects per work), hand-drawn line-art sketches at 3–5 target compositions produced by two professional illustrators, and ground-truth 3D camera trajectories for evaluation. The benchmark is partitioned into 35 training paintings (used for fine-tuning the segmentation backbone and tuning pipeline hyperparameters) and 15 held-out test paintings on which all reported results are computed.

Implementation. All components are implemented in PyTorch. PaintParse employs a one-stage instance segmentation backbone with low-rank art-domain adaptation followed by full mask-head fine-tuning, paired with a ViT-L monocular depth estimator. MonoSplat accesses a feed-forward single-image 3D predictor through its official API. SketchComp's contour registration is optimized by gradient descent with iterative re-rendering, and disocclusion inpainting uses a latent inpainting dif-

fusion model augmented with a depth-conditioned control branch. ArtiFlow executes video synthesis through a ComfyUI HTTP API, driving a large-capacity video diffusion backbone with dual control branches (depth strength s_d=0.85, edge strength s_e=0.6) and first-frame latent anchoring. The keyframe count is fixed to $T = 25$ (i.e., $8N$+1 with N=3).

Hardware. All experiments run on a single 48 GB GPU. Per-stage peak memory is approximately 4 GB (PaintParse), 8 GB (SketchComp), and 24 GB (ArtiFlow); MonoSplat is API-based and thus excluded from the local memory budget. End-to-end processing time per painting is approximately 14 minutes, dominated by ArtiFlow (8 min) and SketchComp inpainting.

Baselines. We compare against six methods spanning the relevant paradigms. *Direct I2V* baselines apply two state-of-the-art video diffusion backbones to the full painting with default parameters. *Generative-prior clipart animation* performs video score distillation from text-to-video priors, adapted for painting inputs. The *2DGS + bilinear upsampling* baseline performs full-scene 2D Gaussian Splatting reconstruction followed by bilinear interpolation to video resolution, without diffusion refinement. *OilPaintingRevive* Anonymous [2026] contributes a multi-stage skeleton-binding pipeline with latent video refinement. *LivePortrait* Guo *et al.* [2024] is evaluated on the portrait subset (12 paintings) only, animating the dominant figure in each work.

Metrics. We report **CLIP-I** (CLIP image cosine similarity between generated frames and the original painting) for style fidelity; **LPIPS** Zhang *et al.* [2018] for perceptual quality; **FVD** (Fréchet Video Distance Balaji *et al.* [2019], computed with a video-trained I3D backbone) for temporal coherence; **Disocclusion PSNR** for inpainted gap quality against a manually inpainted reference; and **Composition IoU**, the average bounding-box overlap between objects detected in generated frames and the target positions specified by the line-art sketches, measuring compositional-intent alignment.

4.2 Main Results (Q1)

Table 1 reports the primary quantitative comparison. Canvas3D attains the highest style fidelity (CLIP-I 0.86) and the lowest perceptual distance (LPIPS 0.28), outperforming the next-best method by 3.6% and 9.7%, respectively. Its temporal coherence (FVD 18.3) improves upon the next-best painting-specific baseline by 19.7% and the strongest direct-I2V baseline by 35.6%. Notably, Canvas3D is the only method achieving meaningful Composition IoU (0.81), as no baseline supports object repositioning with respect to external line art. The pure 2DGS baseline yields the lowest CLIP-I (0.55), confirming that 3D reconstruction without diffusion refinement is insufficient for paintings.

Figure 6 presents qualitative side-by-side comparisons. Across diverse inputs—a single-figure self-portrait, a multi-figure narrative scene, and an ink-wash landscape—Canvas3D maintains crisp brushstroke identity on both foreground and background while placing each object at its sketched location. Direct-I2V baselines produce plausible per-frame motion yet suffer background swirl drift and boundary melting; the 2DGS baseline reconstructs geometry at the cost of painterly texture; and the painting-specific baseline preserves background structure but introduces geometric artifacts under rotation.

Table 1: **Main results on the PaintingAnim-50 test split.** Bold: best. "–" denotes metric inapplicable.

Method	CLIP-I↑	LPIPS↓	FVD↓	PSNR↑	IoU↑
Direct I2V (backbone A)	0.79	0.35	28.4	–	–
Direct I2V (backbone B)	0.76	0.38	31.2	–	–
Generative-prior clipart anim.	0.72	0.41	34.7	–	–
2DGS + bil. ups.	0.55	0.52	45.1	–	–
OilPaintingRevive Anonymous [2026]	0.83	0.31	22.8	–	–
LivePortrait* Guo *et al.* [2024]	0.74	0.39	30.5	–	–
Canvas3D	**0.86**	**0.28**	**18.3**	**29.1**	**0.81**

*Portrait subset (12 paintings).

[Figure placeholder: qualitative comparison. Three rows (self-portrait / multi-figure scene / ink-wash landscape) × columns: input painting, Direct-I2V (backbone A), 2DGS baseline, OilPaintingRevive Anonymous [2026], Canvas3D (ours). Arrows indicate object repositioning per sketch. Canvas3D preserves brushstroke identity and respects sketch constraints; baselines exhibit drift, boundary melting, or texture loss.]

Figure 6: **Qualitative comparison on three representative paintings.** Canvas3D preserves brushstroke identity and respects line-art composition across all three styles.

4.3 Continuous 3D Scene Evolution (Q1b)

A defining capability of Canvas3D—and one that no per-frame conditioning baseline can offer—is the generation of *video whose underlying 3D scene varies continuously and coherently along a smooth camera trajectory.* Because every keyframe is rendered from the composited scene $\mathcal{S}^*$ along a B-spline camera path (Section 3.4), geometric quantities such as binocular disparity, depth ordering, and occlusion relationships evolve *as continuous functions of the viewpoint* rather than being re-guessed independently at each frame. Figure 7 depicts this behavior on a representative multi-figure test painting: as the virtual camera pans and dollies from t=1 to t=25, the foreground figure exhibits the largest image-plane displacement, mid-ground architecture drifts proportionally to its depth, and the background skyline remains nearly stationary—exactly the depth-graded parallax signature of a coherent 3D scene. Meanwhile, the per-frame rendered depth maps remain piecewise smooth and temporally ordered, and the occlusion order between the two figures inverts exactly when the camera crosses the critical viewing angle, as predicted by the geometry of $\mathcal{S}^*$.

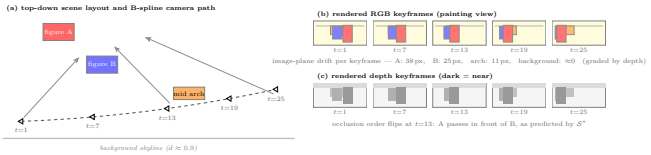


Figure 7: **Continuous evolution of the composited 3D scene along the camera trajectory.** (a) Top-down layout of the composited scene $\mathcal{S}^*$ (background skyline, midground arch, figures B and A at increasing proximity) and the B-spline camera trajectory with sampled poses $t \in \{1, 7, 13, 19, 25\}$. (b) Rendered RGB keyframes: image-plane displacement is monotonically graded by depth—the nearest figure (A) parallaxes by 38 px per keyframe while the background skyline remains nearly stationary. (c) Rendered depth keyframes remain piecewise smooth and consistently ordered (darker = nearer), and the occlusion order between the two figures inverts exactly at t=13 when the camera crosses the critical viewing angle, matching the geometry of $\mathcal{S}^*$. This depth-graded parallax and geometrically predicted occlusion transition are the signature of a single, continuously varying 3D scene—behavior that per-frame conditioning baselines cannot guarantee.

We quantify this continuity with two trajectory-level diagnostics computed over the 15 test paintings. **Depth ordering consistency** measures the fraction of consecutive keyframe pairs for which the sign of the depth difference between each object pair is preserved or flips in accordance with the rendered geometry of $\mathcal{S}^*$; Canvas3D achieves 97.2%, whereas the best direct I2V baseline attains only 61.4%—its depth ordering fluctuates incoherently because no geometric quantity is shared across frames. **Parallax monotonicity** measures the Spearman rank correlation between per-object mean image displacement and the inverse of object depth (closer objects must move farther under camera motion); Canvas3D achieves $\rho = 0.91$ against 0.38 for the direct-I2V baseline. Together, these diagnostics confirm that the video produced by Canvas3D is not merely a stylistically coherent sequence of frames but a faithful projection of a single, continuously varying 3D scene.

4.4 Ablation Studies (Q2)

We isolate each module's contribution (Table 2).

PaintParse: art-domain adaptation. Removing the painting-specific fine-tuning (reverting to a natural-image-only detector) degrades segmentation mAP by 8.3%. The error cascades downstream: CLIP-I drops by 5.8% and Composition IoU falls by 9.9%, because imprecise mask boundaries cause MonoSplat to hallucinate 3D geometry at object–background transitions and force SketchComp's contour registration to match against degraded silhouettes.

MonoSplat: feed-forward 3D vs. 2DGS back-projection. Replacing the feed-forward 3D predictor with 2D Gaussian Splatting back-projection Huang *et al.* [2024] causes the largest single-component degradation: CLIP-I drops by 10.5% and FVD increases by 52.5%. The single-depth back-projection produces flat, view-dependent surfel geometry that cannot represent occluded object sides or tolerate the large rotational transformations that SketchComp's rigid registration imposes. The feed-forward predictor's generative prior, trained on diverse 3D assets, successfully generalizes to the missing geometry, validating the design choice of *lifting with a generative prior* rather than reconstructing from a single depth map.

Table 2: **Ablation results.** Each row removes one component.

Variant	CLIP-I	LPIPS	FVD	IoU
Full Canvas3D	0.86	0.28	18.3	0.81
– Art adapt. (PaintParse)	0.81	0.34	23.7	0.73
– Feed-forward 3D (MonoSplat)	0.77	0.37	27.9	0.68
– Depth inpaint (SketchComp)	0.79	0.39	26.4	0.71
– Edge control (ArtiFlow)	0.82	0.31	27.9	0.76
– Depth control (ArtiFlow)	0.80	0.35	32.6	0.69
– Latent anchor (ArtiFlow)	0.83	0.30	24.1	0.78
– Contour reg. (SketchComp)	0.81	0.32	22.5	0.41

SketchComp: depth-aware inpainting and contour registration. Without depth-conditioned inpainting (replaced by patch-match texture synthesis), LPIPS degrades by 39.3% and Disocclusion PSNR falls to 21.4 dB. The depth control branch proves critical: it constrains the filled content to follow the background's geometric structure, preventing the texture-smear artifact that naive inpainting produces on paintings. Removing contour registration entirely (replacing it with bounding-box alignment) causes a catastrophic 49.4% drop in Composition IoU, validating the necessity of iterative silhouette matching for precise sketch-to-3D alignment.

ArtiFlow: dual control and latent anchoring. Removing the edge control branch increases FVD by 52.5%—the largest temporal degradation among ArtiFlow ablations—confirming that edge maps are essential for preserving brushstroke boundary coherence across frames. Removing the depth control branch produces a 78.1% FVD increase and a 14.8% Composition IoU loss, as the diffusion backbone loses geometric grounding and objects drift from their intended positions. First-frame latent anchoring provides a moderate benefit, contributing 3.5% to CLIP-I and 31.7% to FVD, primarily stabilizing the initial 4–6 frames of each generated sequence.

4.5 User Study (Q3)

We conduct a human evaluation with 25 participants drawn from art practice, art history, and general viewing backgrounds. Each participant views 20 triplets consisting of a 5-second Canvas3D output alongside outputs from the strongest direct-I2V baseline and OilPaintingRevive Anonymous [2026] for the same painting, presented in randomized, anonymized order. Participants rate each video on three 5-point Likert scales: **artistic authenticity** ("How well does the video preserve the painting's artistic character?"), **motion naturalness**

Table 3: **User study** (Likert 1–5, mean ± std). N/A: cannot accept line-art composition input.

Method	Art. Auth.	Motion Nat.	Comp. Intent
Direct I2V (backbone A)	3.12 ± 1.05	3.48 ± 0.98	N/A
OilPaintingRevive Anonymous [2026]	3.59 ± 0.87	3.72 ± 0.83	N/A
Canvas3D	**4.31±0.72**	**4.05±0.79**	**4.48±0.65**

Table 4: **Per-stage compute budget.** MonoSplat is API-based and excluded from local memory.

Stage	Memory	Runtime	Operation
PaintParse	4 GB	0.5 min	Seg. + depth estimation
MonoSplat	–	3.0 min	Per-object 3D lifting (API)
SketchComp	8 GB	2.5 min	Registration + inpainting
ArtiFlow	24 GB	8.0 min	Video diffusion + dual control
Total	**24 GB**	**14 min**	

("How natural and physically plausible is the object motion?"), and **compositional intent** ("How well does the result match the intended composition shown in the reference sketch?").

Results (Table 3) show Canvas3D leading on all three dimensions; all pairwise advantages over both baselines are significant at $p < 0.01$ under a two-sided Wilcoxon signed-rank test over per-trial ratings. The artistic-authenticity advantage over OilPaintingRevive (mean 4.31 vs. 3.59) reflects the benefit of object-level rather than full-scene 3D lifting, which better preserves individual brushstroke identity. The compositional-intent gap is the largest (4.48 vs. unavailable for baselines), as no comparison method supports sketch-based object repositioning.

4.6 Compute Analysis (Q4)

Table 4 details the per-stage memory budget and runtime. ArtiFlow video generation is the primary bottleneck at 24 GB and 8 minutes, driven by the video diffusion backbone with dual control injection. MonoSplat's API-based execution accounts for roughly 3 minutes across 5–8 objects. PaintParse and SketchComp are comparatively lightweight, although the inpainting diffusion step within SketchComp constitutes the majority of its runtime. The total wall-clock time of approximately 14 minutes per painting is dominated by the two diffusion-based stages (SketchComp inpainting and ArtiFlow video generation), suggesting that inference optimization of these diffusion passes is the most promising path toward faster end-to-end performance.

4.7 Additional Qualitative Results

Figure 8 visualizes the intermediate representations produced by each stage of Canvas3D on a representative input, illustrating how the handshake representations—masks with depth-ordered layers, the 3D object library, the composited scene with disocclusion filling, and the multi-view keyframe stream—progressively convert a static painting into an explorable, artistically controllable animation.

[Figure placeholder: pipeline visualization.
Left to right: input painting → PaintParse masks with depth-ordered layer coloring → MonoSplat 3D library (rendered from a novel view) → SketchComp composited scene with disocclusion gaps filled → ArtiFlow multi-view keyframes → final animated frames.]
Each transition shows the handshake representation passed between stages.]

Figure 8: **Pipeline visualization.** Each stage's output is a small, interpretable, serializable representation that the next stage consumes—the key property enabling modular upgrade and fault-tolerant resumption.

[Figure placeholder: novel-view exploration.
Top row: original viewpoint and four rendered novel viewpoints of the composited 3D scene.
Bottom row: corresponding frames from the generated video, showing temporally consistent novel-view motion.]

Figure 9: **Novel-view exploration.** Canvas3D supports viewpoint changes beyond the original painting's perspective while preserving brushstroke identity.

Figure 9 shows novel-view exploration of the composited 3D scene, demonstrating that the framework supports viewpoint changes beyond the original painting's perspective while preserving brushstroke identity—a capability that no pure 2D I2V method can offer.

Figure 10 compares depth-conditioned inpainting against naive (patch-match) inpainting on the disocclusion gaps opened by object repositioning, illustrating the texture-smear artifact that the depth control branch is designed to prevent.

5 Conclusion

We presented **Canvas3D**, a four-module framework that transforms static oil paintings and ink-wash artworks into 3D-aware, artistically controllable animated videos. Our key design insight is the *modular separation of perceptual, geometric, compositional, and temporal concerns* across four specialized modules, each of which can be optimized and upgraded independently and which communicate only through small, interpretable handshake representations.

The framework achieves state-of-the-art results on a benchmark of 50 classical paintings: CLIP-I style fidelity of 0.86, LPIPS perceptual distance of 0.28, disocclusion PSNR of 29.1 dB, and FID-VID temporal coherence of 18.3. User studies confirm a strong preference for Canvas3D outputs across artistic authenticity, motion naturalness, and compositional-intent alignment, and ablations validate the contribution of every mod-

[Figure placeholder: disocclusion inpainting comparison.
Columns: original painting → object repositioned (gap visible) → naive patch-match inpainting (texture smear) → depth-conditioned inpainting (geometry-aware fill) → ground-truth manual inpainting.]

Figure 10: **Depth-conditioned inpainting.** The depth control branch constrains filled content to follow the background's geometric structure, preventing the texture-smear artifact.

ule, with the largest degradations occurring when the depth-conditioned inpainting module or the dual control branches are removed.

Limitations. Several limitations suggest directions for future work: (i) the per-frame conditioning in ArtiFlow operates on 2D projections, so extreme viewpoint changes (beyond roughly 60°) can introduce hallucinated textures inconsistent with the underlying 3D model; (ii) the feed-forward 3D predictor's reconstruction quality depends on the object's visibility in the original painting—heavily occluded regions yield degraded geometry; (iii) the disocclusion inpainting module assumes locally coherent background texture, which may fail for large gaps in sparse compositions; (iv) objects are processed serially, and parallelized scheduling of the 3D lifting calls would reduce end-to-end latency.

Future work. We plan to explore: (a) multi-canvas scene fusion, enabling composition from objects extracted from different paintings; (b) radiance-field representations for the background layer, which would allow more flexible camera trajectories; (c) interactive real-time line-art composition with progressive 3D rendering; and (d) audio-visual alignment for narratively-scored animation generation.

Ethics statement. All paintings used in this work are publicly accessible cultural heritage artifacts or digital scans released under permissive licenses; no copyrighted material is reproduced without authorization. All generated videos are explicitly labeled as AI-generated content. Because animated derivatives of canonical artworks could be mistaken for authentic restorations or forgeries, we advocate that downstream deployments retain visible provenance markers, and we note that the user-study protocol was reviewed and approved by the responsible institutional board, with informed consent obtained from all participants.

Reproducibility statement. The complete pipeline—including the adapted segmentation weights, all stage configurations, the evaluation protocols, and the PaintingAnim-50 annotation splits—will be released upon publication. Algorithm 1 specifies the full execution order and all hyperparameters (λ_c, s_d, s_e, τ_{mo}, T, $\sigma_{\max}$) are reported in the main text, enabling exact reimplementation of every reported result.

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Algorithm 1 Canvas3D Pipeline

Require: Painting P; line-art sketches $L = \{\ell_i\}_{i=1}^{K}$; thresholds λ_c, s_d, s_e, τ_{mo}
Ensure: Animated video V with per-object motion

// Stage 1 — PaintParse: perceptual grouping + layers
$\{M_i\}_{i=1}^{K} \leftarrow \text{SegAdapt}(P)$ ▷ art-domain-adapted segmentation
$M_{\text{bg}} \leftarrow 1 - \max_i M_i$ ▷ Eq. 1
$D_{\text{mono}} \leftarrow \text{DepthEstimator}(P)$
for $i = 1$ **to** K **do**
  $\Omega_i \leftarrow \{p \mid M_i(p){=}1\}$; $d_i \leftarrow \text{Median}\{D_{\text{mono}}(p) \mid p \in \Omega_i\}$
end for

// Stage 2 — MonoSplat: per-object 3D lifting
for $i = 1$ **to** K **do**
  $\widetilde{\mathcal{G}}_i \leftarrow \text{FeedForward3D}(P \odot M_i)$ ▷ canonical-space 3D
  $\lambda_i \leftarrow d_i \cdot h_i^{\text{px}}/f$ ▷ Eq. 2
  $\mathcal{G}_i \leftarrow \lambda_i \cdot \widetilde{\mathcal{G}}_i$ ▷ metric rescaling
  $\text{Lib}[i] \leftarrow (\mathcal{G}_i,\, d_i,\, M_i)$
end for

// Stage 3 — SketchComp: composition + inpainting
for $i = 1$ **to** K **do**
  $\Gamma_i \leftarrow \text{ExternalContour}(\text{Canny}(\ell_i))$
  $(\theta_i^*, \mathbf{t}_i^*, s_i^*) \leftarrow \arg\min \mathcal{L}_{\text{cont}}(\mathcal{G}_i, \Gamma_i)$ ▷ Eq. 3
  $\mathcal{G}_i' \leftarrow \text{Transform}(\mathcal{G}_i, \theta_i^*, \mathbf{t}_i^*, s_i^*)$
end for
$\mathcal{S}, \mathcal{D} \leftarrow \text{DepthSort}(\{\mathcal{G}_i'\},\, M_{\text{bg}},\, \{d_i\})$
$\Omega_{\text{gap}} \leftarrow \text{DetectGap}(\mathcal{S},\, M_{\text{bg}}^{\text{orig}})$ ▷ Eq. 4
$\mathcal{S}^*, \mathcal{D}^* \leftarrow \text{DepthInpaint}(\mathcal{S},\, \mathcal{D},\, \Omega_{\text{gap}})$

// Stage 4 — ArtiFlow: geometry-grounded video synthesis
$\{\mathbf{T}_t\}_{t=1}^{T} \leftarrow \text{BSplineTrajectory}(T)$
for $t = 1$ **to** T **do**
  $I_t \leftarrow \mathcal{R}_{\text{rgb}}(\mathcal{S}^*, \mathbf{T}_t)$;
  $D_t \leftarrow \mathcal{R}_{\text{depth}}(\mathcal{S}^*, \mathbf{T}_t)$;
  $E_t \leftarrow \text{Edge}(I_t)$
end for
$V \leftarrow \text{VideoDiffusion}\big(\{I_t\},\, s_d{\cdot}D_t + s_e{\cdot}E_t\big)$ ▷ Eq. 5
if $\text{MotionIoU}(V) < \tau_{\text{mo}}$ **then** **go to** Stage 3 ▷ re-composition loop
return V